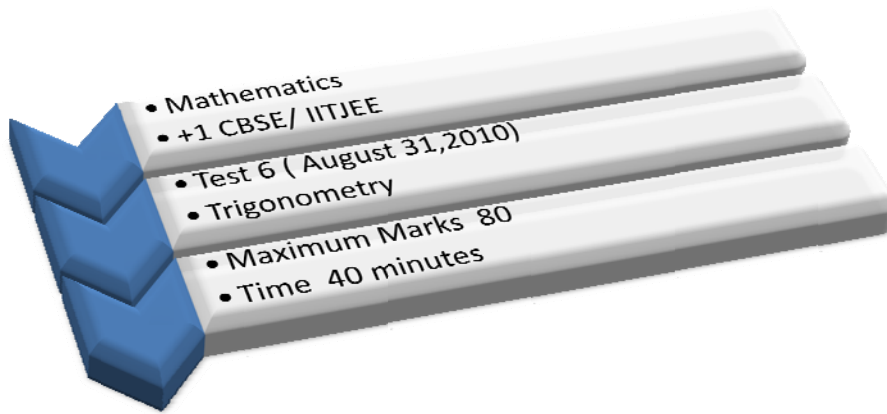




Name
Class
R. No.



Instructions

1. The test contains three sections.
2. The first section is measure of angle, second is Trigonometric Identities and third is heights and distances.

Section I

Q1: The number of sides in two regular polygons are as 5 : 4, and the difference between their angles is 9° ; find the number of sides in the polygons.

10 Marks

Q2: When the moon is setting at any given place the angle that is subtended at its centre by the radius of the earth passing through the given place is $57'$. If the earth's radius be 3960 miles, find approximately the distance of the moon.

10 Marks

Section II

Q3: Prove the following :

30 Marks

$$i) \left(\frac{1}{\sec^2 \alpha - \cos^2 \alpha} + \frac{1}{\cos^2 \alpha - \sin^2 \alpha} \right) \cos^2 \alpha \sin^2 \alpha = \frac{1 - \cos^2 \alpha \sin^2 \alpha}{2 + \cos^2 \alpha \sin^2 \alpha}$$

ii) $(\tan \alpha + \operatorname{cosec} \beta)^2 - (\cot \beta - \sec \alpha)^2 = 2 \tan \alpha \cot \beta (\operatorname{cosec} \alpha + \sec \beta)$

Section III

Q4: A man observes that at a point due south of a certain tower its angle of elevation is 60° ; he then walks 300 feet due west on a horizontal plane and finds that the angle of elevation is then 30° ; find the height of the tower and his original distance from it.

15 Marks

Q5: A man stands at a point X on the bank XY of a river with straight and parallel banks and observes that the line joining X to a point Z on the opposite bank makes an angle of 30° with XY . He then goes along the bank a distance of 200 yards to Y and finds that the angle ZYX is 60° . Find the breadth of the river.

15 Marks